

Curriculum Vitae

Alexander Lerch
Associate Dean for Research & Creative Practice
Professor
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Georgia Institute of Technology



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Google Scholar	scholar.google.com/citations?user=29dF3UIAAAAJ

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1 Curriculum Vitae

1.1 Education

Ph.D.	Technical University of Berlin, Germany Major (summa cum laude): Audio Communication Thesis: <i>Software-Based Extraction of Objective Parameters from Music Performances</i>	2004–2008
Dipl.-Ing.	Technical University of Berlin, Germany Major (summa cum laude): Electrical Engineering Thesis: <i>Qualitätsbeurteilung von codierten Audiosignalen mittels eines objektiven Verfahrens (An objective approach to quality evaluation of encoded audio signals)</i>	1994–2000
w/o degree	University of Arts Berlin, Germany Major: Tonmeister (Music Production)	1996–2000

1.2 Employment History

Professor	School of Music Georgia Institute of Technology, USA	2025–pres.
Associate Professor	School of Music Georgia Institute of Technology, USA	2019–2025.
Assistant Professor	School of Music Georgia Institute of Technology, USA	2013–2019
CEO & Head of Research	zplane.development GmbH & Co KG Berlin, Germany	2001–2013

1.2.1 Administrative Appointments

Associate Dean for Research & Creative Practice	College of Design Georgia Institute of Technology, USA	2023–pres.
Director of Graduate Studies	School of Music Georgia Institute of Technology, USA	2021–2023

1.2.2 Temporary and Part-time Appointments

Adjunct Professor	School of Electrical and Computer Engineering Georgia Institute of Technology, USA	01/2023–pres.
Visiting Professor	Music AI Institute Central Conservatory of Music Beijing, China	06–07/2023
Visiting Professor	Institute of Cultural and Creative Industry Shanghai Jiao Tong University, China	06–07/2019
Visiting Professor	School of Information Science & Technology ShanghaiTech University, China	07/2018
Lecturer (part-time)	Technical University & University of Arts Berlin, Germany	2002–2013
Visiting Scholar (Post-Doc)	Computer Science Department University of Victoria, Canada	2010

1.3 Talks & Presentations

1.3.1 Invited Talks

1. Alexander Lerch, “Music AI — Threats and Opportunities,” Keynote Speaker: *Symposium on AI in Music: Stakeholder perspectives from practitioners, industry and academia*, virtual, 2025.
2. Alexander Lerch, “From Audio to Music Understanding — Machine Learning for Music Analysis, Processing, & Generation,” Speaker: *Technical University Ilmenau*, virtual, 2025.
3. Alexander Lerch, “Music Information Retrieval,” Speaker: *AI4Opt Faculty Training Program*, Georgia Institute of Technology, 2025.
4. Alexander Lerch, “Artificial Intelligence and Music,” Lecture: *Rotary Club of Gwinnett County*, Atlanta, 2024.
5. Alexander Lerch, “The Data Challenge in Music Information Retrieval,” Keynote Speaker: *27th International Symposium on Frontiers of Research in Speech and Music (FRSM)*, Surat, India, 2023.
6. Alexander Lerch, “From Entrepreneur to Academia,” Speaker: *The Sound of AI Accelerator*, virtual, 2023.
7. Alexander Lerch, “Music Information Retrieval,” Speaker: *Santa Cruz Artificial Intelligence Interest Group*, virtual, 2023.
8. Alexander Lerch, “Audio Classification with Insufficient Data,” Speaker: *Central Conservatory of Music, Beijing*, virtual, 2022.
9. Alexander Lerch, “Controllable Music Generation,” Keynote Speaker: *Summit on Music Intelligence*, virtual, 2021.
10. Alexander Lerch, “Musical Audio Analysis with Insufficient Data,” Speaker: *ADASP Seminar, Telecom ParisTech*, virtual, 2021.
11. Alexander Lerch, “Audio Content Analysis,” Speaker: *Applications of Machine Learning in Audio, Audio Engineering Society event*, virtual, 2020.
12. Alexander Lerch, “Music Information Retrieval. An Introduction,” Speaker: *University of Pennsylvania*, Philadelphia, US, 2020.
13. Alexander Lerch, “Constrained Music Generation — User Control of Generative Parameters,” Speaker: *Electronic Arts*, Redwood City, US, 2019.
14. Alexander Lerch, “Research Directions at the Music Informatics Group — Music Information Retrieval, Audio Analysis, & Music Generation,” Speaker: *Netflix*, Los Gatos, US, 2019.
15. Alexander Lerch, “Research Directions at the Music Informatics Group — Music Information

- Retrieval, Audio Analysis, & Music Generation,” Speaker: *Bytedance*, Palo Alto, US, 2019.
16. Alexander Lerch, “Research Directions at the Music Informatics Group — Music Information Retrieval, Audio Analysis, & Music Generation,” Speaker: *Gracenote*, Emeryville, US, 2019.
 17. Alexander Lerch, “Research Directions at the Music Informatics Group — Music Information Retrieval, Audio Analysis, & Music Generation,” Speaker: *Dolby Labs*, San Francisco, US, 2019.
 18. Alexander Lerch, “Music & Artificial Intelligence: Current Directions and Outlook,” Speaker: *Shanghai International Entertainment Law Conference*, Shanghai, China, 2019.
 19. Alexander Lerch, “Music Technology: Overview and Current Research,” Speaker: *Music Beijing — 28th China International Exhibition on Pro Audio, Light, Music & Technology*, Beijing, China, 2019.
 20. Alexander Lerch, “Music Performance Analysis and Automatic Assessment,” Speaker: *Universitat Pompeu Fabra (UPF)*, Barcelona, Spain, 2018.
 21. Alexander Lerch, “Music Information Retrieval — On the Data Challenge in Music Transcription,” Speaker: *Peking University*, Beijing, China, 2018.
 22. Alexander Lerch, “Music Information Retrieval — On the Data Challenge in Music Transcription,” Speaker: *Alibaba Music*, Hangzhou, China, 2018.
 23. Alexander Lerch, “Commercialization of Emerging Technology — Music Technology and Music Information Retrieval,” Speaker: *Shanghai Forum 2018*, Shanghai, China, 2018.
 24. Alexander Lerch, “Music Information Retrieval — Introduction, Technology, and Applications,” Speaker: *Shanghai Conservatory of Music*, Shanghai, China, 2018.
 25. Alexander Lerch, “Music Information Retrieval — Introduction, Technology, and Applications,” Speaker: *School of Music, Shanghai University*, Shanghai, China, 2018.
 26. Alexander Lerch, “Drum Transcription and Sample Detection with Non-Negative Matrix Factorization,” Keynote: *Berlin MIR Meetup*, Berlin, Germany, 2017.
 27. Alexander Lerch, “Commercialization of Emerging Technology — Music Technology and Music Information Retrieval,” Speaker: *Xi’an International Forum on Tech Mining and Tech Transfer*, Xi’an, China, 2017.
 28. Alexander Lerch and Gil Weinberg, “Machine Learning & Music,” Workshop: *Moogfest*, Durham, NC, 2017.
 29. Alexander Lerch and Stefan Weinzierl, “Music Performance Analysis,” Tutorial Speaker: *AES Conference on Semantic Audio*, Erlangen, Germany, 2017.
 30. Alexander Lerch, “Listening Machines — An Introduction to Music Information Retrieval,” Workshop: *SASE Southeast Regional Conference*, Atlanta, GA, 2017.
 31. Alexander Lerch, “Music Information Retrieval Research at Georgia Tech,” Keynote: *Audio Communication Day, Technical University of Berlin*, Berlin, Germany, 2016.
 32. Alexander Lerch, “Music Information Retrieval,” Speaker: *Polytecnico di Bari*, Bari, Italy, 2016.
 33. Alexander Lerch, “Music Performance Analysis,” Speaker: *University of Potsdam*, Potsdam, Germany, 2016.
 34. Alexander Lerch, “Academia in Berlin,” Speaker: *DFG Forum: Research in Germany (Destination Europe)*, Atlanta, GA, 2014.
 35. Alexander Lerch, “Applications in Signal Processing and Music Informatics,” Speaker: *Louisiana State University*, Baton Rouge, LA, 2013.
 36. Alexander Lerch, “Music Technology for Music Production, Broadcast & DJing,” Speaker: *Centre for Digital Music, Queen Mary University of London*, UK, 2012.

1.3.2 Session & Panel Chairing

1. Alexander Lerch, Session Chair: *International Society for Music Information Retrieval Conference (ISMIR)*, Milan, Italy, 2023.
2. Alexander Lerch, “Multitrack,” Session Chair: *International Society for Music Information Retrieval*

Conference (ISMIR), Suzhou, China, 2017.

3. Alexander Lerch, “Automatic Music Transcription,” Session Chair: *AES Conference on Semantic Audio*, Erlangen, Germany, 2017.
4. Alexander Lerch, “Industry Panel,” Panel Chair: *International Conference on Music Information Retrieval (ISMIR)*, Malaga, Spain, 2015.

1.3.3 Panels

1. Stephan Baumann, Matthias Krebs, Joachim Gossmann, Alexander Lerch, “Impact of AI,” Panel: *Symposium on AI in Music: Stakeholder perspectives from practitioners, industry and academia*, virtual, 2025.
2. Roger Dannenberg, Zhiyao Duan, Anssi Klapuri, Alexander Lerch, Daniel Ray, “MIR for Music Education,” Panel: *International Society for Music Conference*, virtual, 2021.
3. Gil Weinberg, Jason Freeman, Alexander Lerch, Michael Winters, and Takahiko Tsuchiya, “Research at Georgia Tech Center for Music Technology (GTCMT),” Panel: *Moogfest*, Durham, NC, 2017.
4. Markus Lemmens, Thomas Arend, Dirk Kanngießer, Alexander Meyer-Scharenberg, Grzegorz Garbacz, Nina Lemmens, Oliver Günther, and Alexander Lerch, “Entrepreneurship Praktisch,” Panel: *German Academic International Network (GAIN) annual meeting*, San Francisco, CA, 2017.
5. Haoshu Peng, Alexander Lerch, Nils Newman, and Luo Lino, “Motivating Technology Transfer from Research and Development to Commercialization,” Panel: *Pujiang Innovation Forum*, Shanghai, China, 2017.
6. Christian Uhle, Alexander Lerch, Josh Reiss, Udo Zölzer, “Intelligent Digital Audio Effects,” Panel: *126th AES Convention*, Munich, Germany, 2009.

1.4 Honors & Awards

1.4.1 Fellowships

1. *Post-Doctorate Fellowship*
Deutscher Akademischer Austausch-Dienst (German Academic Exchange Service), 2010.
2. *NaFög PhD Fellowship*
City of Berlin, 2005–2007.

1.4.2 Awards

1. *Python Notebooks for Audio Content Analysis*
 Sponsor: Georgia Tech Open Source Program Office
 Role: PI
 Collaborators: n/a
 Total Amount: \$8,200
 Period: Jan. 2025 – Jun. 2025
2. *Unrestricted Gift*
 Sponsor: Netflix Inc
 Role: PI
 Collaborators: n/a
 Total Amount: \$30,000
 Period: Jan. 2023

3. *REU: Research Experience for Undergraduates*

Sponsor: National Science Foundation (NSF)
 Role: Co-PI
 Collaborators: Subhrajit Guhathakurta (PI)
 Total Amount: \$16,000
 Period: Jan. 2024 – Dec. 2024

4. *EAGER: Urban Sensing of Pedestrians through Integrated, Cost-Effective, and Scalable Audio Sensor Networks*

Sponsor: National Science Foundation (NSF)
 Role: Co-PI
 Collaborators: Subhrajit Guhathakurta (PI)
 Total Amount: \$299,999
 Period: Apr. 2022 – Mar. 2024

5. *Music Instrument Recognition*

Sponsor: Gracenote, Inc.
 Role: PI
 Collaborators: n/a
 Total Amount: \$179,530
 Period: Aug. 2016 – Dec. 2020

6. *Music Performance Assessment*

Sponsor: Microsoft, Corp.
 Role: PI
 Collaborators: n/a
 Total Amount: \$7,500
 Period: Mar. 2020 – Jun. 2020

7. *Auditory Event Recognition*

Sponsor: SmartEar, Inc.
 Role: PI
 Collaborators: James Hays (Co-PI)
 Total Amount: \$30,000
 Period: Aug. 2017 – May 2018

8. *GT-FIRE: Interactive real-time assessment for music ensembles*

Sponsor: Georgia Institute of Technology
 Role: Co-PI
 Collaborators: T. Hsu (PI)
 Total Amount: \$40,000
 Period: Jul. 2015 – Jun. 2017

9. *Faculty Development Grant: Performance-specific models for music information retrieval*

Sponsor: College of Architecture, Georgia Institute of Technology
 Role: PI
 Collaborators: n/a
 Total Amount: \$10,942
 Period: Jan. 2016 – Jul. 2016

1.4.3 Other Awards1. *Hesburgh Award Teaching Fellow*

Center for Teaching & Learning, Georgia Institute of Technology, 2019–2020.

2. *Teaching with Technology Partnership Faculty Fellow*

Center for Teaching & Learning, Georgia Institute of Technology, 2017–2018.

3. *Class of 1969 Teaching Fellow*

Center for the Enhancement of Teaching & Learning, Georgia Institute of Technology, 2014–2015.

1.5 Teaching & Advising

1.5.1 Courses Taught

Course Number and Title	Term
MUSI 6201/4457: Comp. Music Analysis/Audio Content Analysis	2021, 2019, 2017, 2015, 2013 Fall
MUSI 6202/4459: Music Digital Signal Processing	2023, 2019, 2017, 2015 Spring
MUSI 6106: Audio Software Engineering	2022, 2020, 2018, 2016, 2014 Spring
MUSI 7100: Music Tech Research Lab	2013–pres. Spring and Fall
MUSI 4699: Undergraduate Research	2021 Spring, 2020 Spring and Fall, 2019 Fall, 2018 Spring, 2017 Fall
MUSI 4705: Music Tech Capstone	2022 Spring, 2021 Spring and Fall, 2020 Fall, 2019 Spring
MUSI 3770: Project Studio Technology	2020 Fall
MUSI 2525: Intro to Audio Technology I	2018, 2016 Fall
MUSI 8903: Deep Learning for Music	2018 Fall
MUSI 8001: Research Methods	2018, 2015 Fall
MUSI 6001: Music Perception, Cognition	2014 Fall
Before at TU Berlin & UdK Berlin: Acoustics, Digital Audio Effects, Digital Audio Technology, Audio Content Analysis	

1.5.2 Course & curriculum development

1. New Degree Proposal: BS in Music Technology, 2015.
2. New Course Development:
 - *MUSI 6106: Audio Software Engineering*, 2014–2020.
 - *MUSI 2525: Introduction to Audio Technology I*, 2016.
 - *Audio Content Analysis*, 2009–2021.

1.5.3 Advising & Mentoring

- Advised Students at Georgia Tech
 - 6 PhD students (5 graduated)
 - 62 MS students
 - 11 BS students
- Mentoring

- IEEE SPS SigMA mentoring program, IEEE, 2025–pres.
- New-to-ISMIR paper mentoring program, ISMIR, 2023–pres.
- Early Career Faculty mentoring program, GT, 2019–pres.
- Women in Music Information Retrieval, ISMIR 2016–pres.
- IEEE Young Professionals, IEEE, 2018.

1.5.4 Supervised theses

1. Ashvala Vinay, School of Music, Georgia Institute of Technology, 2024.
PhD Thesis: *"Towards Perceptual Metrics for Audio Quality Assessment"*
Principal Advisor(s): Alexander Lerch
2. Ashis Pati, School of Music, Georgia Institute of Technology, 2020.
PhD Thesis: *"Learning to Manipulate Latent Representations of Deep Generative Models"*
Principal Advisor(s): Alexander Lerch
3. Siddharth Gururani, School of Music, Georgia Institute of Technology, 2020.
PhD Thesis: *"Weakly Supervised Learning for Musical Instrument Classification"*
Principal Advisor(s): Alexander Lerch
4. Chih-Wei Wu, School of Music, Georgia Institute of Technology, 2018.
PhD Thesis: *"Addressing the data challenge in automatic drum transcription with labeled and unlabeled data"*
Principal Advisor(s): Alexander Lerch
5. Fabian Seipel, Audio Communications Group, Technical University of Berlin, 2018.
MS Thesis: *"Musical instrument identification using convolutional neural networks"*
Principal Advisor(s): Alexander Lerch
6. Udit Gupta, ECE, Georgia Institute of Technology, 2017.
PhD Thesis: *"Towards development of no-reference objective measures for perceptual evaluation of singing voice separation"*
Principal Advisor(s): Elliot Moore and Alexander Lerch
7. Vinod Subramanian, Birla Institute of Technology & Science, Pilani, IN, 2016.
BS Thesis: *"Music segmentation with rhythmic descriptors"*
Principal Advisor(s): K.K. Gupta and Alexander Lerch
8. Cian O'Brian, School of Music, Georgia Institute of Technology, 2015.
MS Thesis: *"Supervised Feature Learning via Sparse Coding for Music Information Retrieval"*
Principal Advisor(s): Alexander Lerch
9. Athanasios Lykartsis, Audio Communications Group, Technical University of Berlin, 2014.
MS Thesis: *"Evaluation of Accent-Based Rhythmic Descriptors for Genre Classification of Musical Signals"*
Principal Advisor(s): Alexander Lerch
10. Patricio López Serrano Erickson, Audio Communications Group, Technical University of Berlin, 2014.
MS Thesis: *"Audio Thumbnailing: Structural Segmentation and its Applications to Popular Music"*
Principal Advisor(s): Alexander Lerch
11. Henrik von Coler, Audio Communications Group, Technical University of Berlin, 2013.
MS Thesis: *"Blind Note-Level Segmentation of Monophonic Music Using Hidden Markov Models"*
Principal Advisor(s): Alexander Lerch
12. Constantin Wiesener, Audio Communications Group, Technical University of Berlin, 2010.
MS Thesis: *"Adaptive Verfahren der Audiorestauration für Echtzeitanwendungen (Adaptive Methods for Real-time Audio Signal Restoration)"*
Principal Advisor(s): Stefan Weinzierl and Alexander Lerch

13. Holger Kirchhoff, Audio Communications Group, Technical University of Berlin, 2008.
MS Thesis: *"Synchronisation von Audio Signalen (Alignment of Audio Signals)"*
Principal Advisor(s): *Stefan Weinzierl and Alexander Lerch*
14. Bernardo Escalona-Espinoza, Audio Communications Group, Technical University of Berlin, 2008.
MS Thesis: *"Downbeat and Meter Estimation in Audio Signals"*
Principal Advisor(s): *Stefan Weinzierl and Alexander Lerch*
15. Noam Yogev, Audio Communications Group, Technical University of Berlin, 2008.
MS Thesis: *"Realisierung eines Systems zur automatischen Harmonisierung (Realization of a System for Automatic Harmonization)"*
Principal Advisor(s): *Stefan Weinzierl and Alexander Lerch*
16. Ingmar-Leander Klich, Audio Communications Group, Technical University of Berlin, 2004.
MS Thesis: *"Automatische Erkennung von Onsets in Musiksignalen zur Steuerung von Beattracking-Systemen (Automatic Onset Detection in Music Signals for Controlling Beat Tracking Systems)"*
Principal Advisor(s): *Stefan Weinzierl and Alexander Lerch*
17. Juan-José Burred, Telecommunications Group, Technical University of Berlin, 2003.
MS Thesis: *"An Objective Approach to Content-Based Audio Signal Classification"*
Principal Advisor(s): *Thomas Sikora and Alexander Lerch*

1.6 Service

1.6.1 Service on thesis committees

1. Farshad Jafari, Georgia Institute of Technology, 2025.
MS Thesis: *"Multimodal Film Music Dataset"*
Principal Advisor(s): *Claire Arthur*
2. Venkatakrishnan Vaidyanathapuram Krishnan, Georgia Institute of Technology, 2025.
MS Thesis: *"The Perception of Phase Intercept Distortion and its Relevance to Audio Machine Learning"*
Principal Advisor(s): *Nathaniel Condit-Schultz*
3. Qianyi (Rose) Sun, Georgia Institute of Technology, 2024.
MS Thesis: *"Interpretable Quantitative Evaluation Metrics Of Generative Models In Symbolic Music"*
Principal Advisor(s): *Claire Arthur*
4. Aran Komatsuzaki, Georgia Institute of Technology, 2023.
PhD Thesis: *"Improving Foundation Models"*
Principal Advisor(s): *Heinrich Matzinger*
5. Sichen Wang, Georgia Institute of Technology, 2021.
PhD Thesis: *"An Indirect Speech Enhancement Framework through Intermediate Noisy Speech Targets"*
Principal Advisor(s): *Chin-Hui Lee*
6. Ashvala Vinay, Georgia Institute of Technology, 2020.
MS Thesis: *"The Sound within: Learning Audio Features from Electroencephalogram Recordings of Audio Perception"*
Principal Advisor(s): *Grace Leslie*
7. Rong Gong, Universitat Pompeu Fabra, Barcelona, 2018.
PhD Thesis: *"Automatic Assessment of Singing Voice Pronunciation: A Case Study with Jingju Music"*
Principal Advisor(s): *Xavier Serra*
8. Juan Carlos Martinez-Nieto, Georgia Institute of Technology, 2018.
PhD Thesis: *"Algorithmic Score Language: Extending Common Western Music Notation for Representing Logical Behaviors"*
Principal Advisor(s): *Jason Freeman*

9. Lamtharn Hantrakul, Georgia Institute of Technology, 2018.
MS Thesis: *"Regressing Dexterous Finger Flexions Using Machine Learning and Multi-Channel Signal Element Ultrasound Transducers"*
Principal Advisor(s): Gil Weinberg
10. Mason Bretan, Georgia Institute of Technology, 2017.
PhD Thesis: *"Towards an Embodied Musical Mind: Generative Algorithms for Robotic Musicians"*
Principal Advisor(s): Gil Weinberg
11. Deepak Edakkattil Gopinath, Georgia Institute of Technology, 2015.
MS Thesis: *"Increasing Expressivity in Robotic Drummers"*
Principal Advisor(s): Gil Weinberg
12. Nashlie Sephus, Georgia Institute of Technology, 2014.
PhD Thesis: *"A Framework for Exploiting Modulation Spectral Features in Music Data Mining and Other Applications"*
Principal Advisor(s): Aaron Lanterman and David Anderson
13. Asif Ali, Georgia Institute of Technology, 2013.
PhD Thesis: *"Voice Query-By-Example using an Ergodic Hidden Markov Model of Speech"*
Principal Advisor(s): Mark Clements

1.6.2 Conference Organization

1. *Workshop Co-Chair*, Music Interfaces for Listening and Creation Workshop (MILC) at IUI Conference, 2023.
2. *Conference General Co-Chair*, International Society for Music Information Retrieval Conference (ISMIR), 2021.
3. *Steering Committee*, Music Demixing Workshop (MDX), 2021.
4. *Conference Co-Chair*, Web Audio Conference (WAC), 2016.

1.6.3 Editorships and reviewing

1. *Journals*:
 - *Associate Editor*, "Open Journal of Signal Processing," IEEE, 2024–pres.
 - *Guest Editor*, Special Issue "Machine Learning Applied to Music/ Audio Signal Processing," MDPI Electronics, 2021.
 - *Reviewer*: Applied Sciences (MDPI), EURASIP Journal on Audio, Speech, and Music Processing (Springer), Journal of Selected Topics in Signal Processing (IEEE), Journal of the Audio Engineering Society (AES), Journal of the Electronics and Telecommunications Research Institute (ETRI), Journal of New Music Research (Routledge), Machine Learning (Springer), Neural Computing and Applications (Springer), PLOS ONE, Sensors (MDPI), Transactions of the International Society for Music Information Retrieval (TISMIR), Transactions on Audio, Speech and Language Processing (IEEE), Transactions on Intelligent Systems and Technology (ACM).
2. *Conferences*:
 - *Area Chair/Meta Reviewer/Program Committee*: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), International Society for Music Information Retrieval Conference (ISMIR)
 - *Reviewer*: ACM Multimedia (ACMMM), AES Conference on Semantic Audio, European Signal Processing Conference (EUSIPCO), International Conference on Machine Learning (ICML), International Joint Conference on Artificial Intelligence (IJCAI), International Joint

Conference on Neural Networks (IJCNN), International Society of Music Information Retrieval Conference (ISMIR), International Symposium on the Internet of Sounds (IS2), Web Audio Conference (WAC), Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE), IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA).

3. *Funding Agencies*: FWF (Austria), DFG (Germany), NSERC (Canada), FFG (Austria).
4. *Tenure Cases*: University of Richmond (2025), Oregon State University (2025), Hong Kong University of Science and Technology (2024), New York University (2021), University of Miami (2020).

1.6.4 Committees

- *College of Design AI Workgroup* (Chair), Georgia Institute of Technology, 2025–pres.
- *College of Design Space Study Steering Committee* (Member), Georgia Institute of Technology, 2025–pres.
- *College of Computing Fellowship Selection Committee* (Member), Georgia Institute of Technology, 2025–pres.
- *Search Committee* (Member), Chair of School of Music, Georgia Institute of Technology, 2025–pres.
- *Associate Deans for Research Workgroup* (Member), A2RU, 2024–pres.
- *AI Steering Committee* (Member), Georgia Institute of Technology, 2024–pres.
- *EVPR Institute Research Awards* (Member), Georgia Institute of Technology, 2024–pres.
- *Search for the Executive Director, SPACE Research Institute* (Member), Georgia Institute of Technology, 2024–pres.
- *Space Policy Guidance Working Group* (Member), Georgia Institute of Technology, 2024.
- *Reappointment, Promotion and Tenure Committee* (Member), School of Music, Georgia Institute of Technology, 2023–pres.
- *Internal Research Advisory Board* (Member), Georgia Institute of Technology, 2023–2024
- *International Advisory Committee* (Member), Georgia Institute of Technology, 2019–2022
- *Graduate Faculty Council* (Member), Georgia Institute of Technology, 2019–2022.
- *Reappointment, Promotion and Tenure Committee* (Chair), School of Music, Georgia Institute of Technology, 2019–2023.
- *College Curriculum Committee* (Member), College of Design, Georgia Institute of Technology, 2018–pres.
- *Graduate Admissions Committee* (Chair), Recruiting and Admission, School of Music, Georgia Institute of Technology, 2014–pres.
- *Search Committee* (Chair), Music Tech Tenure-Track, School of Music, Georgia Institute of Technology, 2018–2019.
- *Search Committee* (Member), Visiting 1-year Faculty Position, School of Music, Georgia Institute of Technology, 2018.
- *Search Committee* (Member), School Chair, School of Music, Georgia Institute of Technology, 2017–2018.
- *Search Committee* (Member), Music Tech Tenure-Track, School of Music, Georgia Institute of Technology, 2016–2017.
- *Search Committee* (Member), Conductor Tenure-Track, School of Music, Georgia Institute of Technology, 2013–2014.

1.6.5 Organizations & Societies

- *Member*, IEEE SPS AASP TC (Signal Processing Society Audio and Acoustic Signal Processing Technical Committee), 2026–pres.
- *Board Member*, International Society for Music Information Retrieval (ISMIR), 2026–pres.
- *DAAD Research Ambassador*, Deutscher Akademischer Austausch-Dienst (German Academic Exchange Service), 2014–pres.
- *Advisory Board Member*, GAIN (German Academic International Network), 2016–2020.
- *Senior Member*: Institute of Electrical and Electronics Engineers (IEEE)
- *Member*: International Society for Music Information Retrieval (ISMIR)
- *Member*: Association for Computing Machinery (ACM)

2 Publications

2.1 Books

- [1] A. Lerch, *An Introduction to Audio Content Analysis: Music Information Retrieval Tasks and Applications*, 2nd ed. Hoboken, N.J.: Wiley-IEEE Press, 2023, ISBN: 978-1-119-89094-2.
- [2] A. Lerch, *An Introduction to Audio Content Analysis: Applications in Signal Processing and Music Informatics*. Hoboken: Wiley-IEEE Press, 2012, ISBN: 978-1-118-26682-3.
- [3] A. Lerch, *Software-Based Extraction of Objective Parameters from Music Performances*. München: GRIN Verlag, 2009, ISBN: 978-3-640-29496-1.

2.2 Book Chapters

- [4] U. Zölzer and A. Lerch, “Digitale Audio-Effekte,” in *Handbuch der Audiotechnik*, S. Weinzierl, Ed., 2nd ed., Berlin, Heidelberg: Springer, 2025, pp. 1–22, ISBN: 978-3-662-60357-4. DOI: [10.1007/978-3-662-60357-4_28-1](https://doi.org/10.1007/978-3-662-60357-4_28-1).
- [5] A. Lerch, “Audioinhaltsanalyse,” in *Handbuch der Audiotechnik*, S. Weinzierl, Ed., 2nd ed., Berlin, Heidelberg: Springer Berlin Heidelberg, 2023, pp. 1–20, ISBN: 978-3-662-60357-4. DOI: [10.1007/978-3-662-60357-4_8-1](https://doi.org/10.1007/978-3-662-60357-4_8-1).
- [6] A. Lerch, “Grundlagen digitaler Audiosignale,” in *Handbuch der Audiotechnik*, S. Weinzierl, Ed., 2nd ed., Berlin, Heidelberg: Springer Berlin Heidelberg, 2023, pp. 1–13, ISBN: 978-3-662-60357-4.
- [7] J. Herre, S. Disch, and A. Lerch, “Quellcodierung,” in *Handbuch der Audiotechnik*, S. Weinzierl, Ed., 2nd ed., Berlin, Heidelberg: Springer, 2022, pp. 1–23, ISBN: 978-3-662-60357-4. DOI: [10.1007/978-3-662-60357-4_34-1](https://doi.org/10.1007/978-3-662-60357-4_34-1).
- [8] A. Lerch, “The Relation Between Music Technology and Music Industry,” in *Springer Handbook of Systematic Musicology*, ser. Springer Handbooks, R. Bader, Ed., Springer, Berlin, Heidelberg, 2018, pp. 899–909, ISBN: 978-3-662-55004-5. DOI: [10.1007/978-3-662-55004-5_44](https://doi.org/10.1007/978-3-662-55004-5_44).
- [9] A. Lerch, “Music Information Retrieval,” in *Akustische Grundlagen der Musik*, ser. Handbuch der Systematischen Musikwissenschaft 5, S. Weinzierl, Ed., Laaber, 2014, pp. 79–102, ISBN: 978-3-89007-699-7.
- [10] A. Lerch, “Software-gestützte Merkmalsextraktion für die musikalische Aufführungsanalyse,” in *Gemessene Interpretation - Computergestützte Aufführungsanalyse im Kreuzverhör der Disziplinen*, ser. Klang und Begriff, H. von Loesch and S. Weinzierl, Eds., Mainz: Schott, 2011, pp. 205–212, ISBN: 978-3-7957-0771-2.
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